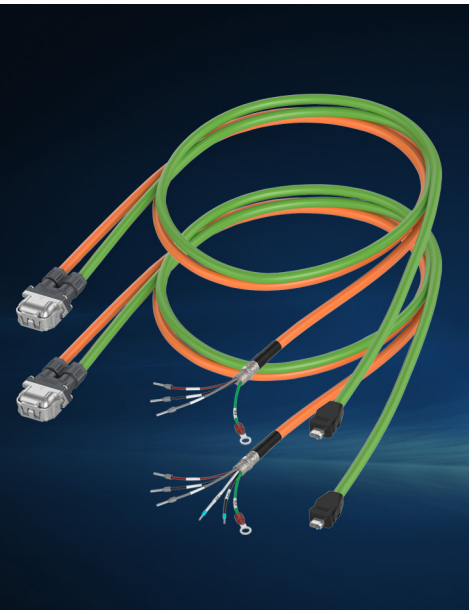
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4/2 **MOTION-CONNECT 350 and 380**
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4/3 **Pre-assembled dual cables for power and communication for SINAMICS S200**
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4/4 **Pre-assembled power cables for SINAMICS S200**
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4/5 **Pre-assembled signal cables for SINAMICS S200**
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4/6 **Pre-assembled brake cables for SINAMICS S200**
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4/6 Technical specifications

4/7 **Connectors for SINAMICS S200**
4/7 Overview

For **selection and ordering data** please refer to section "System overview" "SINAMICS S200 servo drive system" from page 1/16.

Detailed technical information on SINAMICS S200 is available on the internet at: www.siemens.com/sinamics-s200/documentation

In addition, the Siemens Product Configurator can be used on the internet at the following address: www.siemens.com/sinamics-s200/configuration

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MOTION-CONNECT 350 and 380

Overview

MOTION-CONNECT 350 and 380 connection systems

The cables are available in lengths of up to 30 m and one step further optimized in easy and secure installation with "one-push-lock" motor connecting solution for motor shaft heights 20, 30, 40 and 40 Compact.

- MOTION-CONNECT 350 cables, which provide a connection solution for fixed installation applications
- MOTION-CONNECT 380 cables, which provide a connection solution for drag chain installation applications that require a high number of bending cycles

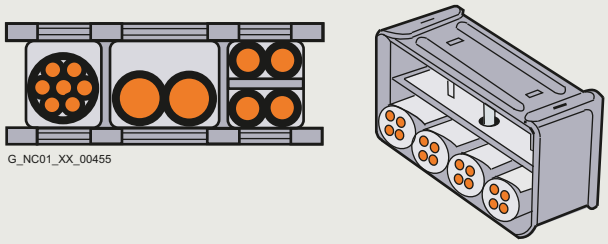
The use of pre-assembled MOTION-CONNECT 350/380 cables ensures high quality and system-tested, problem-free operation.

Degree of protection of pre-assembled power and signal cables is IP65 when closed and connected unless otherwise stated.

MOTION-CONNECT 350/380 cables are not suitable for outdoor use.

MOTION-CONNECT 350/380 cables are approved for a maximum horizontal travel distance of 5 m without support.

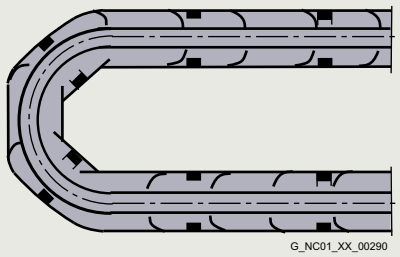
The cables must be unwound without twisting.



To maximize the service life of the cable carrier and cables, cables in the carrier made from different materials must be separated in the cable carrier using spacers. The spacers must be filled evenly to ensure that the position of the cables does not change during operation. The cables should be distributed as symmetrically as possible according to their weights and dimensions. Cables with different outer diameters should be separated by spacers as well.

When inserting pre-assembled cables into the cable carrier, do not pull at the connector, as this may damage the strain relief or cable clamping.

The cables must not be fixed in the cable carrier.
They must be freely movable.



The cables must be able to be moved without applying force, specifically in the bending radii of the carrier.
The specified minimum bending radii must be adhered to.

The cable fixings must be attached at both ends at an appropriate distance from the end points of the moving parts in a dead zone.

Cables must be installed in accordance with the instructions supplied by the cable carrier manufacturer.

In case of vibration load and with horizontal or vertical cable entries, we recommend that the cable is additionally fixed if between the cable strain relief on the cable carrier and the terminal at the motor part of the cable is hanging loose or is not routed. To prevent machine vibrations being transmitted to the connectors, the cable should be fixed at the moving part where the motor is mounted.

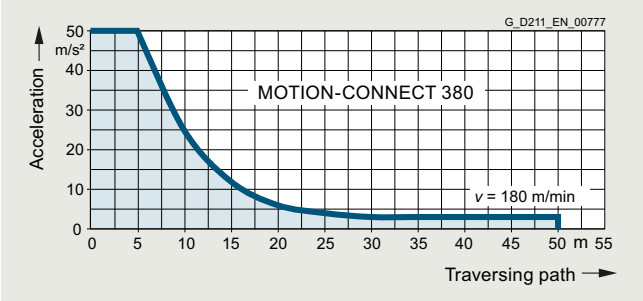
Derating factors for power and signal cables

Ambient air temperature °C (°F)	Derating factor according to EN 60204-1 Table D.1
30 (86)	1.10
35 (95)	1.05
40 (104)	1.00
45 (113)	0.96
50 (122)	0.90
55 (131)	0.84
60 (140)	0.78

Characteristic curves

Characteristic curve for MOTION-CONNECT 380

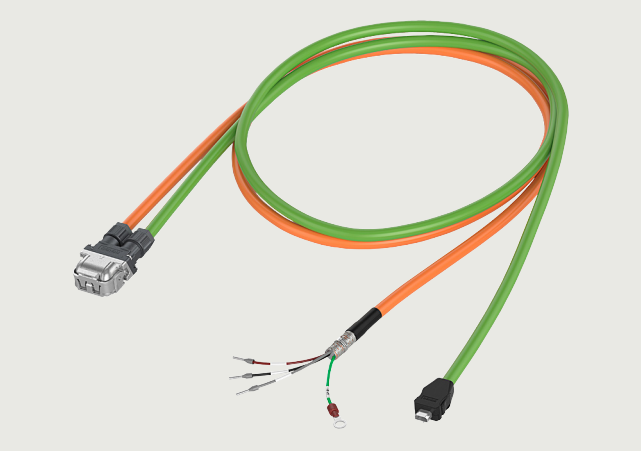
The blue area beneath the characteristic curve represents the potential range of use for the cables. The characteristic curves represent the tested operating points.



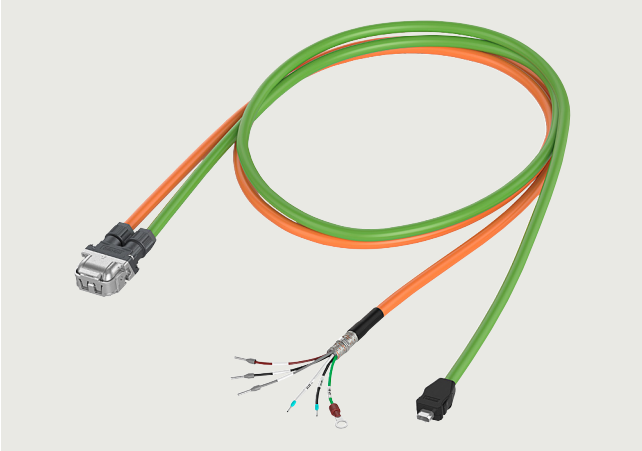
Permissible acceleration for MOTION?CONNECT 380 signal and power cables

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Overview



Example: MOTION-CONNECT 350/380, dual cable without brake cores for SIMOTICS S-1FL2 Low Inertia/Medium Inertia servomotors



Example: MOTION-CONNECT 350/380, dual cable with brake cores for SIMOTICS S-1FL2 Low Inertia/Medium Inertia servomotors

For selection and ordering data please refer to section "System overview" "SINAMICS S200 servo drive system" from page 1/16.

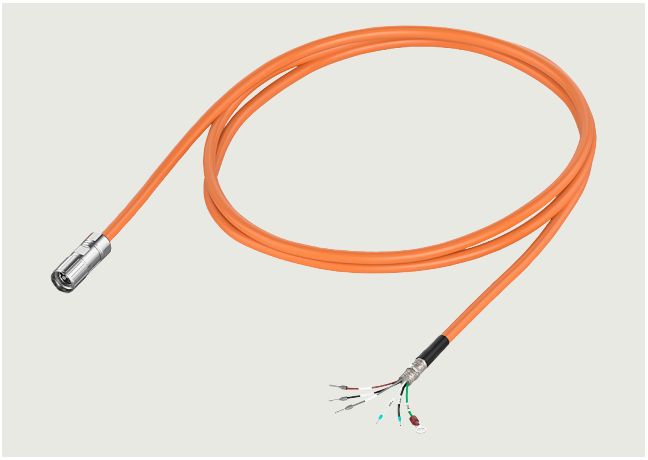
Technical specifications

Dual cable	**without brake cores**		**with brake cores**	
Product name	**MOTION-CONNECT 350**	**MOTION-CONNECT 380**	**MOTION-CONNECT 350**	**MOTION-CONNECT 380**
• for coverter	• SINAMICS S200 200 ... 240 V 1 AC / 3 AC and • SINAMICS S200 380 ... 480 V 3 AC			
• for motor	• SIMOTICS S-1FL2 Low Inertia, 0.05 kW ... 1 kW and • SIMOTICS S-1FL2 Medium Inertia, 0.2 kW ... 1 kW			
Type	6FX3502-7CD01-.... 6FX3502-7CN01-....	6FX3802-7CD01-.... 6FX3802-7CN01-....	6FX3502-7DD01-.... 6FX3502-7DN01-....	6FX3802-7DD01-.... 6FX3802-7DN01-....
No. of cores	10	10	12	12
Degree of protection motor side (when mated and locked)	IP65	IP65	IP65	IP65
Certificate of suitability				
• RoHS	Yes	Yes	Yes	Yes
• CE and UKCA	Yes	Yes	Yes	Yes
• cURus	Yes	Yes	Yes	Yes
Rated voltage U_0/U	300 V/1000 V	300 V/1000 V	300 V/1000 V	300 V/1000 V
Test voltage, rms	4 kV	4 kV	4 kV	4 kV
Operating temperature on the surface				
• Fixed installation	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)
• Flexible installation	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)
Tensile stress, max.				
• Fixed installation	50 N/mm²	50 N/mm²	50 N/mm²	50 N/mm²
• Flexible installation	20 N/mm²	20 N/mm²	20 N/mm²	20 N/mm²
Smallest bending radius				
• Fixed installation	4 × diameter	4 × diameter	4 × diameter	4 × diameter
• Flexible installation	7.5 × diameter	7.5 × diameter	7.5 × diameter	7.5 × diameter
Torsional stress	Absolute 30°/m	Absolute 30°/m	Absolute 30°/m	Absolute 30°/m
Bending	1 million	5 million	1 million	5 million
Insulation material	PP	PP	PP	PP
Oil resistance	EN 60811-2-1	EN 60811-2-1	EN 60811-2-1	EN 60811-2-1
Outer jacket	PVC	PUR	PVC	PUR
Outer diameter				
• Power cable	7.5 mm ±0.2 mm	7.5 mm ±0.2 mm	7.5 mm ±0.2 mm	7.5 mm ±0.2 mm
• Signal cable	7.0 mm ±0.2 mm	7.0 mm ±0.2 mm	7.0 mm ±0.2 mm	7.0 mm ±0.2 mm
Flame-retardant	EN 60332-1-1 to 1-3	EN 60332-1-1 to 1-3	EN 60332-1-1 to 1-3	EN 60332-1-1 to 1-3

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Pre-assembled power cables for SINAMICS S200

Overview



Example: MOTION-CONNECT 350/380, power cable with brake cores for SIMOTICS S-1FL2 Low Inertia/Medium Inertia servomotors



Example: MOTION-CONNECT 350/380, power cable without brake cores for SIMOTICS S-1FL2 High Inertia servomotors

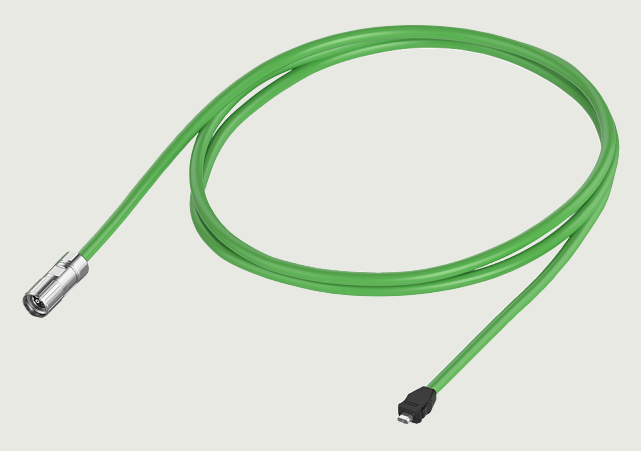
For selection and ordering data please refer to section "System overview" "SINAMICS S200 servo drive system" from page 1/16.

Technical specifications

Power cable	**without brake cores**				**with brake cores**	
Product name	**MOTION-CONNECT 350**	**MOTION-CONNECT 380**	**MOTION-CONNECT 350**	**MOTION-CONNECT 380**	**MOTION-CONNECT 350**	**MOTION-CONNECT 380**
• for coverter	• SINAMICS S200 380 ... 480 V 3 AC					
• for motor	• SIMOTICS S-1FL2 Low Inertia, 1.5 kW... 2 kW and • SIMOTICS S-1FL2 Medium Inertia, 1 kW ... 1.5 kW		• SIMOTICS S-1FL2 High Inertia, 0.4 kW ... 7 kW		• SIMOTICS S-1FL2 Low Inertia, 1.5 kW ... 2 kW and • SIMOTICS S-1FL2 Medium Inertia, 1 kW ... 1.5 kW	
Type	6FX3502-5CK32-....	6FX3802-5CK32-....	6FX3502-5CL01-.... 6FX3502-5CL02-.... 6FX3502-5CL12-....	6FX3802-5CL01-.... 6FX3802-5CL02-.... 6FX3802-5CL12-....	6FX3502-5DK32-....	6FX3802-5DK32-....
No. of cores	4	4	4	4	6	6
Degree of protection motor side (when mated and locked)	IP65	IP65	IP65	IP65	IP65	IP65
Certificate of suitability						
• RoHS	Yes	Yes	Yes	Yes	Yes	Yes
• CE and UKCA	Yes	Yes	Yes	Yes	Yes	Yes
• cURus	Yes	Yes	Yes	Yes	Yes	Yes
Rated voltage U_0/U	600 V/1000 V	600 V/1000 V	600 V/1000 V	600 V/1000 V	600 V/1000 V	600 V/1000 V
Test voltage, rms	4 kV	4 kV	4 kV	4 kV	4 kV	4 kV
Operating temperature on the surface						
• Fixed installation	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)
• Flexible installation	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)
Tensile stress, max.						
• Fixed installation	50 N/mm²	50 N/mm²	50 N/mm²	50 N/mm²	50 N/mm²	50 N/mm²
• Flexible installation	20 N/mm²	20 N/mm²	20 N/mm²	20 N/mm²	20 N/mm²	20 N/mm²
Smallest bending radius						
• Fixed installation	4 × diameter	4 × diameter	4 × diameter	4 × diameter	4 × diameter	4 × diameter
• Flexible installation	7.5 × diameter	7.5 × diameter	7.5 × diameter	7.5 × diameter	7.5 × diameter	7.5 × diameter
Torsional stress	Absolute 30°/m	Absolute 30°/m	Absolute 30°/m	Absolute 30°/m	Absolute 30°/m	Absolute 30°/m
Bending	1 million	5 million	1 million	5 million	1 million	5 million
Insulation material	PP	PP	PP	PP	PP	PP
Oil resistance	EN 60811-2-1	EN 60811-2-1	EN 60811-2-1	EN 60811-2-1	EN 60811-2-1	EN 60811-2-1
Outer jacket	PVC	PUR	PVC	PUR	PVC	PUR
Outer diameter	7.5 mm ±0.2 mm	7.5 mm ±0.2 mm	6FX3.02-5CL01-.... : 7.5 mm ±0.2 mm 6FX3.02-5CL02-.... : 7.8 mm ±0.3 mm 6FX3.02-5CL12-.... : 9.0 mm ±0.3 mm		7.5 mm ±0.2 mm	7.5 mm ±0.2 mm
Flame-retardant	EN 60332-1-1 to 1-3	EN 60332-1-1 to 1-3	EN 60332-1-1 to 1-3	EN 60332-1-1 to 1-3	EN 60332-1-1 to 1-3	EN 60332-1-1 to 1-3

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Overview



Example: MOTION-CONNECT 350/380, signal cable for encoder connection for SIMOTICS S-1FL2 Medium Inertia servomotors



Example: MOTION-CONNECT 350/380, signal cable for encoder connection for SIMOTICS S-1FL2 High Inertia servomotors

For selection and ordering data please refer to section "System overview" "SINAMICS S200 servo drive system" from page 1/16.

Technical specifications

Signal cable	**for encoder connection**			
Product name	**MOTION-CONNECT 350**	**MOTION-CONNECT 380**	**MOTION-CONNECT 350**	**MOTION-CONNECT 380**
• for coverter	• SINAMICS S200 380 ... 480 V 3 AC			
• for motor	• SIMOTICS S-1FL2 Low Inertia, 1.5 kW ... 2 kW and • SIMOTICS S-1FL2 Medium Inertia, 1 kW ... 1.5 kW		• SIMOTICS S-1FL2 High Inertia, 0.4 kW ... 7 kW	
Type	6FX3502-2XB11-....	6FX3802-2XB11-....	6FX3502-2XB12-....	6FX3802-2XB12-....
No. of cores	6	6	6	6
Degree of protection motor side (when mated and locked)	IP65	IP65	IP65	IP65
Certificate of suitability				
• RoHS	Yes	Yes	Yes	Yes
• CE and UKCA	Yes	Yes	Yes	Yes
• cURus	Yes	Yes	Yes	Yes
Rated voltage U_0/U	30 V/300 V	30 V/300 V	30 V/300 V	30 V/300 V
Test voltage, rms	500 V	500 V	500 V	500 V
Operating temperature on the surface				
• Fixed installation	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)
• Flexible installation	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)
Tensile stress, max.				
• Fixed installation	50 N/mm²	50 N/mm²	50 N/mm²	50 N/mm²
• Flexible installation	20 N/mm²	20 N/mm²	20 N/mm²	20 N/mm²
Smallest bending radius				
• Fixed installation	4 × diameter	4 × diameter	4 × diameter	4 × diameter
• Flexible installation	7.5 × diameter	7.5 × diameter	7.5 × diameter	7.5 × diameter
Torsional stress	Absolute 30°/m	Absolute 30°/m	Absolute 30°/m	Absolute 30°/m
Bending	1 million	5 million	1 million	5 million
Insulation material	PP	PP	PP	PP
Oil resistance	EN 60811-2-1	EN 60811-2-1	EN 60811-2-1	EN 60811-2-1
Outer jacket	PVC	PUR	PVC	PUR
Outer diameter	7.0 mm ±0.2 mm	7.0 mm ±0.2 mm	7.0 mm ±0.2 mm	7.0 mm ±0.2 mm
Flame-retardant	EN 60332-1-1 to 1-3	EN 60332-1-1 to 1-3	EN 60332-1-1 to 1-3	EN 60332-1-1 to 1-3

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Pre-assembled brake cables for SINAMICS S200

Overview



Example: MOTION-CONNECT 350/380, brake cable for brake connection for SIMOTICS S-1FL2 High Inertia servomotors

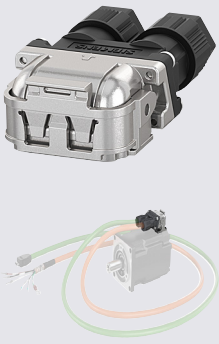
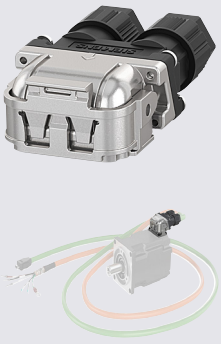
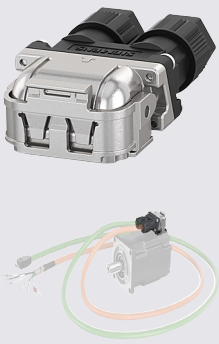
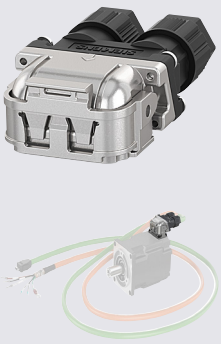
For selection and ordering data please refer to section "System overview" "SINAMICS S200 servo drive system" from page 1/16.

Technical specifications

Brake cable	**for brake connection**	
Product name	**MOTION-CONNECT 350**	**MOTION-CONNECT 380**
• for coverter	• SINAMICS S200 380 … 480 V 3 AC	
• for motor	• SIMOTICS S-1FL2 High Inertia, 0.4 kW … 7 kW	
Type	6FX3502-5BL03-....	6FX3802-5BL03-....
No. of cores	2	2
Degree of protection motor side (when mated and locked)	IP65	IP65
Certificate of suitability		
• RoHS	Yes	Yes
• CE and UKCA	Yes	Yes
• cURus	Yes	Yes
Rated voltage U_0/U	30 V/300 V	30 V/300 V
Test voltage, rms	2000 V	2000 V
Operating temperature on the surface		
• Fixed installation	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)
• Flexible installation	-20 … +80 °C (-4 … +176 °F)	-20 … +80 °C (-4 … +176 °F)
Tensile stress, max.		
• Fixed installation	50 N/mm²	50 N/mm²
• Flexible installation	20 N/mm²	20 N/mm²
Smallest bending radius		
• Fixed installation	4 × diameter	4 × diameter
• Flexible installation	7.5× diameter	7.5× diameter
Torsional stress	Absolute 30°/m	Absolute 30°/m
Bending	1 million	5 million
Insulation material	PP	PP
Oil resistance	EN 60811-2-1	EN 60811-2-1
Outer jacket	PVC	PUR
Outer diameter	6.0 mm ±0.2 mm	6.0 mm ±0.2 mm
Flame-retardant	EN 60332-1-1 to 1-3	EN 60332-1-1 to 1-3

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Overview

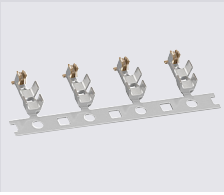
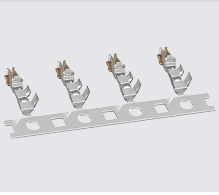
Shaft height	Connectors drive end direction (DE) for hybrid connection	Connectors non drive end direction (NDE) for hybrid connection	Connectors motor side for power connection	for encoder connection	for brake
MOTION-CONNECT connectors for SIMOTICS S-1FL2 Low Inertia servomotors					
20, 30, 40, 40 Compact	6FX2003-3HD00	6FX2003-3HN00	–	–	–
					
52	–	–	6FX2003-0LL63	6FX2003-0DB63	– (in power connector)
					
MOTION-CONNECT connectors for SIMOTICS S-1FL2 Medium Inertia servomotors					
30, 40, 40 Compact	6FX2003-3HD00	6FX2003-3HN00	–	–	–
					
48	–	–	6FX2003-0LL63	6FX2003-0DB63	– (in power connector)
					
MOTION-CONNECT connectors for SIMOTICS S-1FL2 High Inertia servomotors					
45, 65, 90	–	–	6FX2003-0LL13	6FX2003-0DB13	6FX2003-0LL53
					

For selection and ordering data please refer to section "System overview" "SINAMICS S200 servo drive system" from page 1/16.

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Connectors for SINAMICS S200

Overview

	Crimp contacts "power and PE" (2000 pieces per reel) for hybrid connection	Crimp contacts "signal and brake" (2000 pieces per reel) for hybrid connection	Connectors converter side for power connection	IX signal connector, C-coded for encoder connection	for brake
MOTION-CONNECT connectors for SINAMICS S200 servo converter					
	6FX2003-7HP00	6FX2003-7HS00	–	6FX2003-0DE61	–
					

For selection and ordering data please refer to section "System overview" "SINAMICS S200 servo drive system" from page 1/16.